

Fundamentals of Electric Circuits

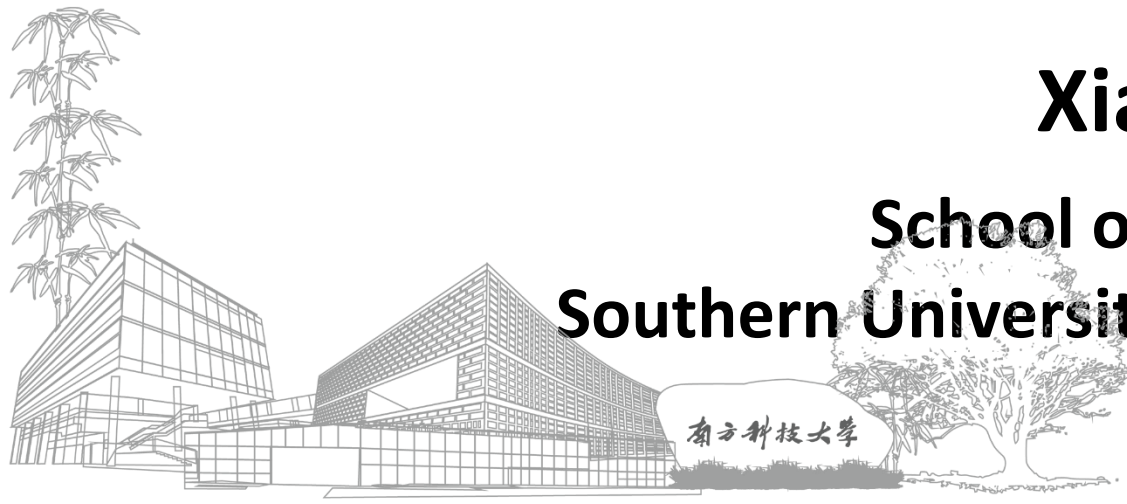
电路基础

Lecture6: Capacitors and Inductors

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Outline

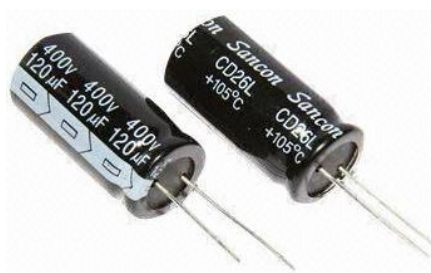
- **Introduction**
- **Capacitors**
- **Series and Parallel Capacitors**
- **Inductors**
- **Series and Parallel Inductors**
- **Applications**

Capacitor (电容) & Inductor (电感)

- We have limited our study to resistive circuits. In this chapter, we shall introduce two new and important passive linear circuit elements: the **capacitor (电容)** and the **inductor (电感)**
 - Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy. For this reason, capacitors and inductors are called **storage elements (储能元件)**
- Available in various values and types



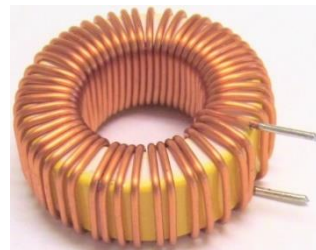
Ceramic (陶瓷) Capacitors



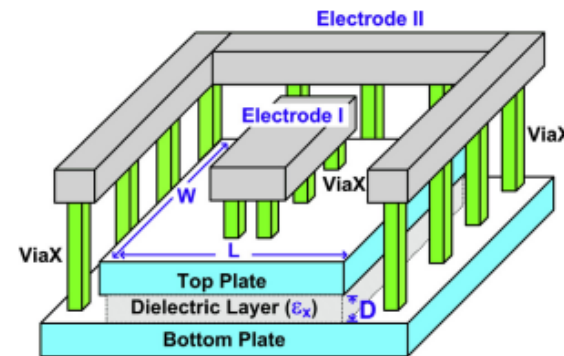
Electrolytic (电解) Capacitor



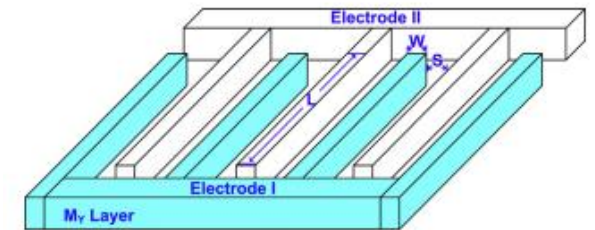
Solenoidal (螺线形) Inductor



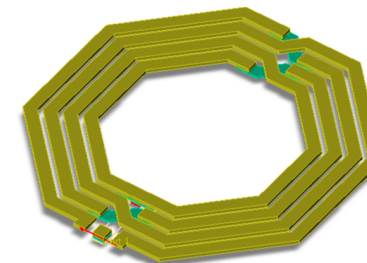
Toroidal (环形) Inductor



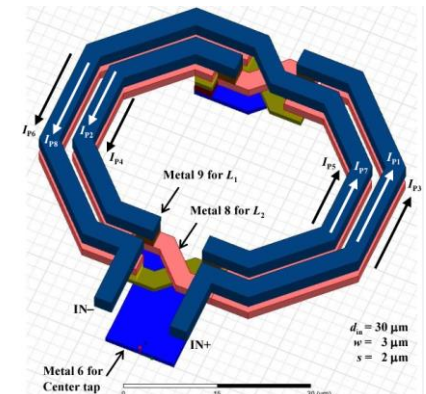
Metal-Insulator-Metal (MIM) capacitors



Metal-Oxide-Metal (MOM) Capacitor

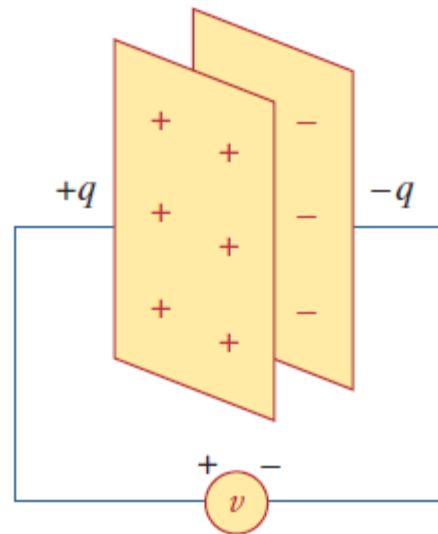
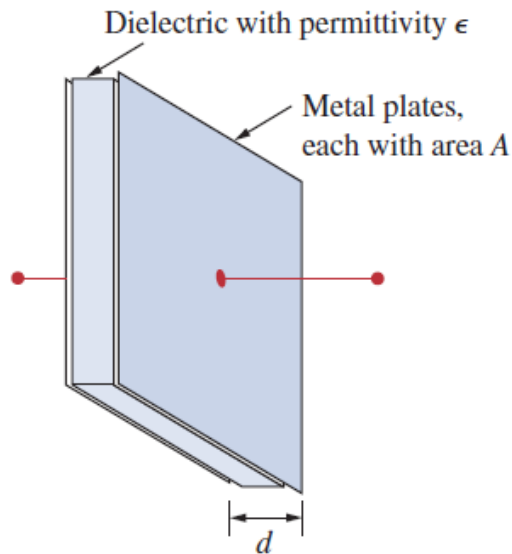


Integrated Inductor



Basics of Capacitor (I)

- A capacitor is a passive element that stores energy in its electric field. Capacitors are used extensively in electronics, communications, computers, and power systems
- It consists of two conducting plates separated by an **insulator (or dielectric, 电介质)**
- When a voltage source v is connected to the capacitor, the source deposits a positive charge $+q$ on one plate and a negative charge $-q$ on the other. The amount of charge is proportional to the applied voltage



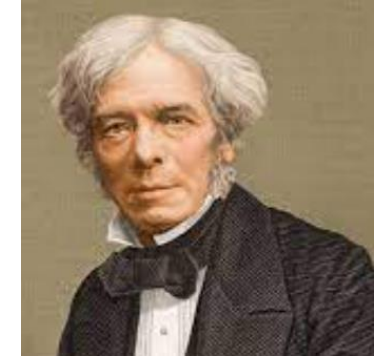
电容器 = “电荷的容器”

$$q = Cv$$

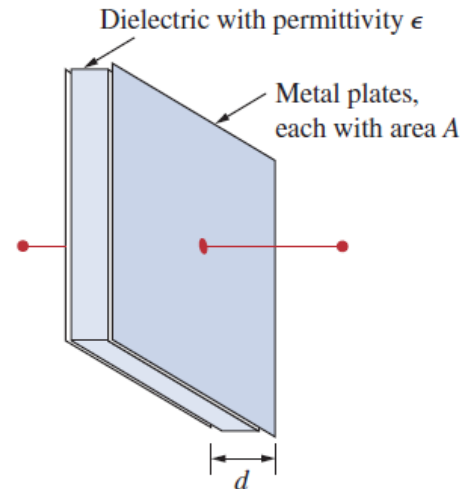
C is the capacitance

Basics of Capacitor (II)

- Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F)
 - Since $q = Cv$, we can find that 1 **farad (法拉)** = 1 coulomb/volt
 - Most capacitors are rated in picofarad (pF) and microfarad (μF)
- We assume the capacitance depends on the physical dimensions of the capacitor, but not on the stored charge q or the applied voltage v . For the parallel-plate capacitor, the capacitance **C** is



Michael Faraday (1791-1867)
English Physicist

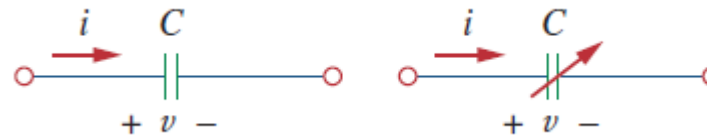


$$C = \frac{\epsilon A}{d}$$

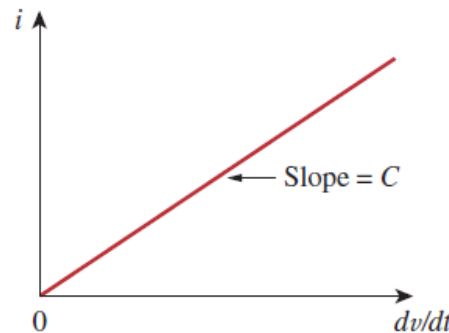
where ϵ is the permittivity of the dielectric (介电常数) material between the plates

V-I Property of Capacitor (I)

- According to the passive sign convention, if $v \cdot i > 0$, the capacitor is being **charged** (充电) (storing energy), and if $v \cdot i < 0$, the capacitor is **discharging** (放电) (providing energy)



- Using the formula for the charge stored in a capacitor, i.e., $q = C \cdot v$, we can find the current-voltage relationship. Take the first derivative with respect to time gives



$$i = \frac{dq}{dt} = C \frac{dv}{dt}$$

Figure 6.6

Current-voltage relationship of a capacitor.

- For a capacitor whose capacitance is independent of the voltage, its current-voltage relationship satisfies this equation and so it is said to be linear

V-I Property of Capacitor (II)

- From the current-voltage relationship, the voltage-current relationship is given by

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

- This shows the capacitor has a memory, which is often exploited in circuits
- The **instantaneous power** (瞬时功率) delivered to the capacitor is

$$p = vi = Cv \frac{dv}{dt}$$

- Therefore, by charging up the capacitor from 0 to q Coulomb, and hence the voltage from 0 to $v = q/C$ volts, the energy stored in the capacitor is

$$w = \int_{t_0}^t Cv' \frac{dv'}{d\tau} d\tau = C \int_0^v v' dv' = \frac{1}{2} Cv^2 = \frac{q^2}{2C}$$

- This represents the energy stored in the electric field that exists between the plates of the capacitor

Characteristics of Capacitors

- A capacitor is an **open circuit to dc** (电容对直流开路)
- The voltage on a capacitor must be continuous, it cannot change abruptly
 - An abrupt change in voltage would require an infinite current which is physically impossible!
- An ideal capacitor does not dissipate energy, meaning stored energy may be retrieved later
- A real capacitor has a parallel-model **leakage resistance** (漏电阻), leading to a slow loss of the stored energy internally
 - This resistance is typically very high, on the order of 100 MΩ and thus can be ignored for many circuit applications

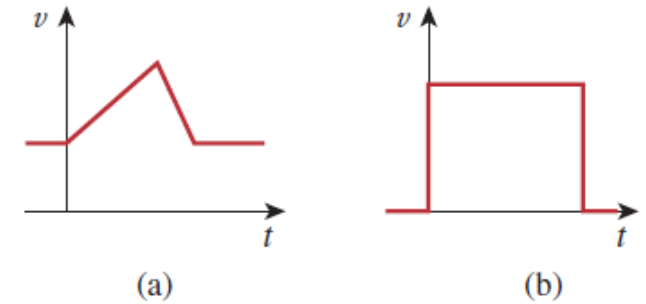


Figure 6.7
Voltage across a capacitor: (a) allowed, (b) not allowable; an abrupt change is not possible.

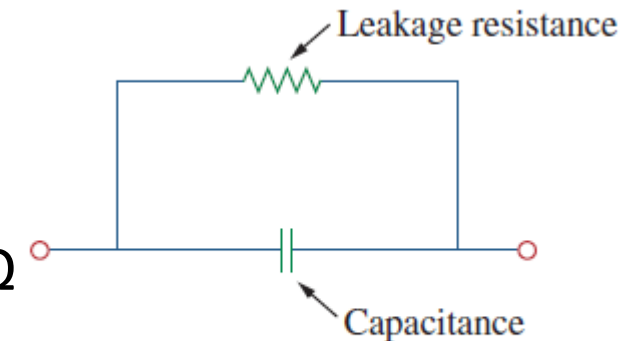
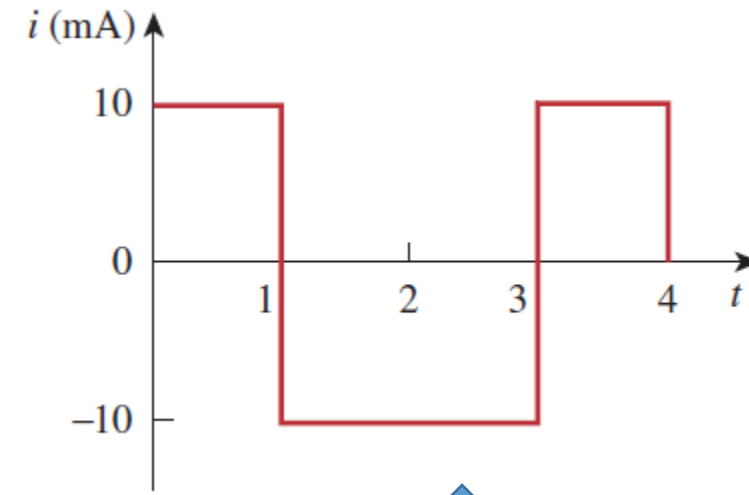
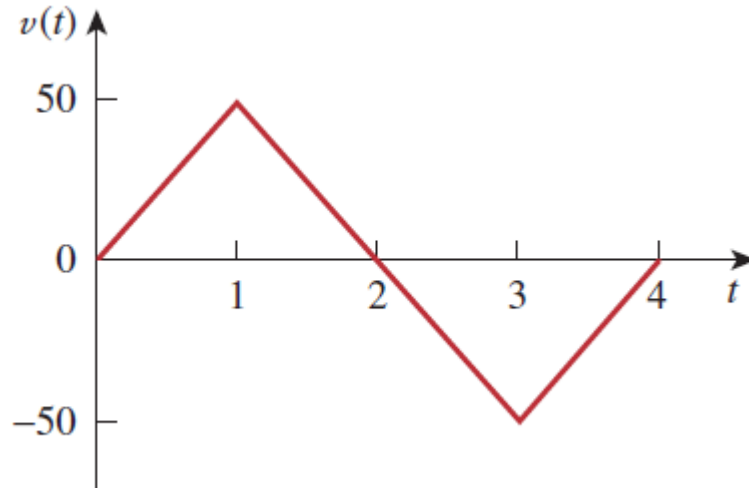


Figure 6.8
Circuit model of a nonideal capacitor.

Example1: Capacitors

- Q: Determine the current through a 200- μF capacitor when it is applied a voltage as shown in the following figure

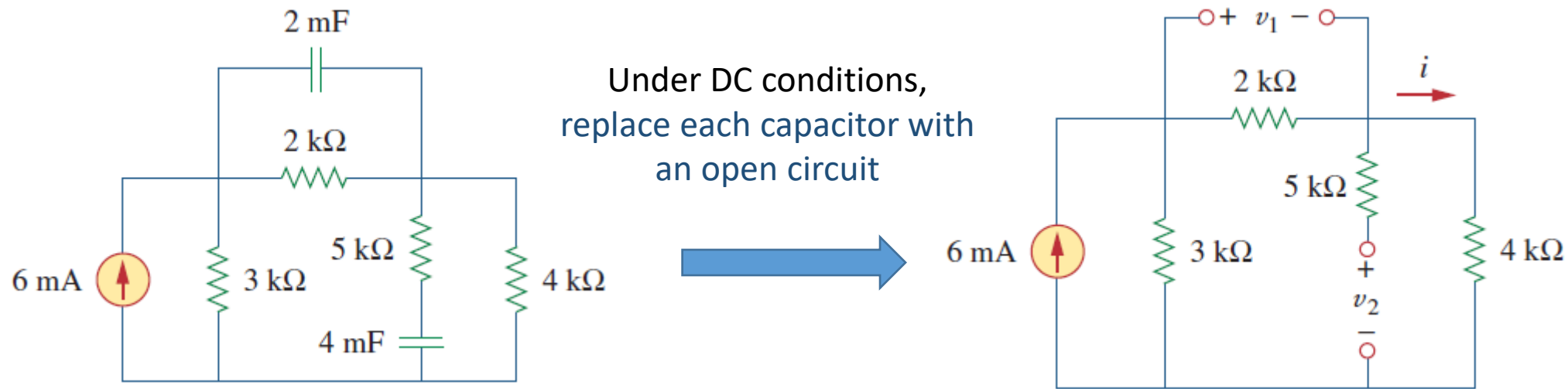


Solution:

$$v(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ 100 - 50t \text{ V} & 1 < t < 3 \\ -200 + 50t \text{ V} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases} \quad i = C \frac{dv}{dt} \quad \longrightarrow \quad i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Example2: Capacitors

- Q: Obtain the energy stored in each capacitor in figure below under DC conditions



Solution:

With only DC source applied for a sufficient long time, capacitors appear as an open-circuit. The figure on the left can then be simplified the one on the right. We have

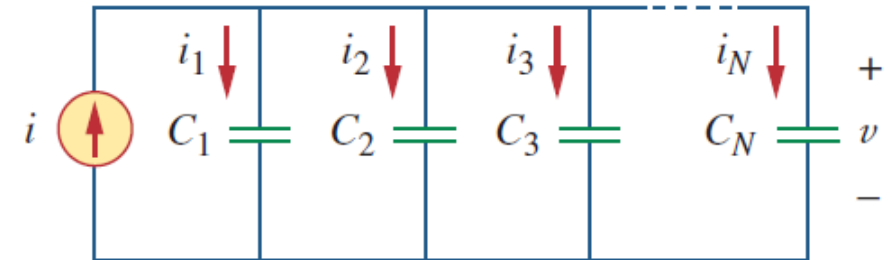
$$v_1 = 2k \cdot [6m \cdot 3k / (3k + 2k + 4k)] = 4 \text{ V}, \text{ so } w_1 = \frac{1}{2} \cdot C_1 \cdot v_1^2 = \frac{1}{2} \cdot 2m \cdot 4^2 = 16 \text{ mJ}$$

$$v_2 = 4k \cdot [6m \cdot 3k / (3k + 2k + 4k)] = 8 \text{ V}, \text{ so } w_2 = \frac{1}{2} \cdot C_2 \cdot v_2^2 = \frac{1}{2} \cdot 4m \cdot 8^2 = 128 \text{ mJ}$$

Parallel Capacitors

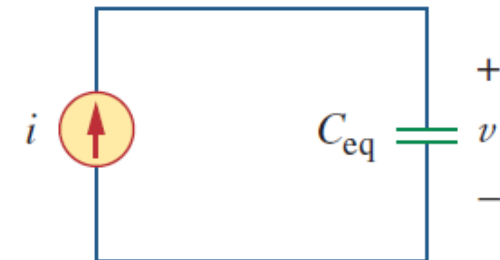
- We learned with resistors that applying the equivalent series and parallel combinations can simplify many circuits
- Starting with N parallel capacitors (并联电容), one can note that the voltages on all the capacitors are the same. Applying KCL

$$i = i_1 + i_2 + i_3 + \cdots + i_N$$



- Taking into consideration the current-voltage relationship of each capacitor

$$\begin{aligned} i &= C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt} \\ &= \left(\sum_{k=1}^N C_k \right) \frac{dv}{dt} = C_{eq} \frac{dv}{dt} \end{aligned}$$



$$C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N$$

Capacitors in parallel combine in the same manner as resistors in series

Series Capacitors

- For a series arrangement of capacitors, each capacitor shares the same current.
Applying KVL to the loop

$$v = v_1 + v_2 + v_3 + \cdots + v_N$$

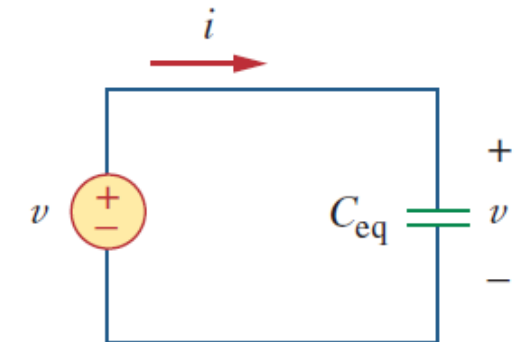
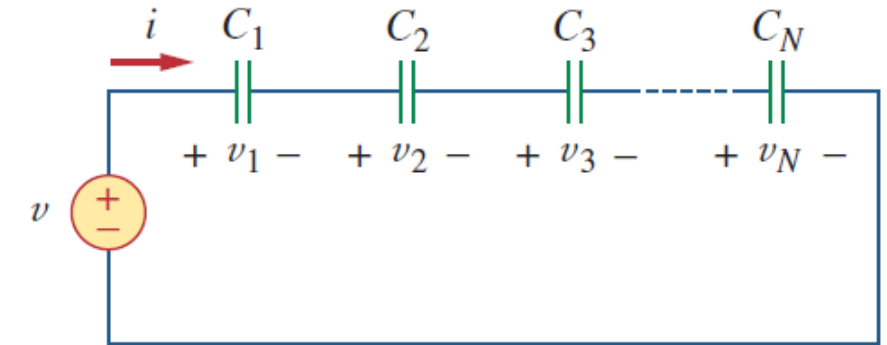
- Now apply the voltage-current relationship

$$v = \frac{1}{C_1} \int_{t_0}^t i(\tau) d\tau + v_1(t_0) + \frac{1}{C_2} \int_{t_0}^t i(\tau) d\tau + v_2(t_0)$$

$$+ \cdots + \frac{1}{C_N} \int_{t_0}^t i(\tau) d\tau + v_N(t_0) = \frac{1}{C_{eq}} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$



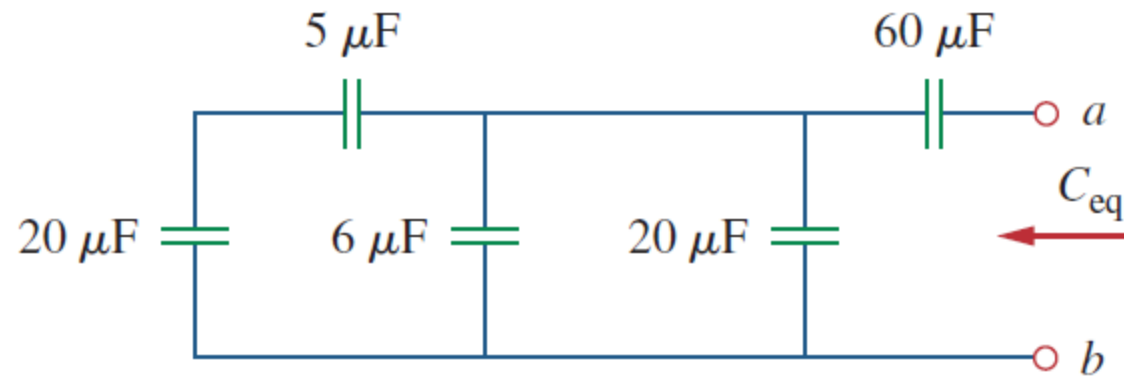
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}$$



- Capacitors in series combine in the same manner as resistors in parallel

Example: Series and Parallel Capacitors

- Q: Find the equivalent capacitance seen between terminals a and b



Solution:

The $20\text{-}\mu\text{F}$ and $5\text{-}\mu\text{F}$ are in series, so their equivalent is $20 \cdot 5 / (20 + 5) = 4\ \mu\text{F}$

This $4\ \mu\text{F}$ is in parallel with the $6\ \mu\text{F}$ and $20\ \mu\text{F}$, so their equivalent is $4 + 6 + 20 = 30\ \mu\text{F}$

Finally, the $60\ \mu\text{F}$ and $30\ \mu\text{F}$ are in series, so $C_{eq} = 30 \cdot 60 / (30 + 60) = 20\ \mu\text{F}$

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- Introduction
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- Series and Parallel Capacitors
- **Inductors**
- **Series and Parallel Inductors**
- Applications

Basics of Inductors

- An **inductor** (电感) is a passive element that stores energy in its magnetic field
- Any conductor of electric current has inductive properties and may be regarded as an inductor. But in order to enhance the inductive effect, a practical inductor is usually formed as a **cylindrical coil** (圆柱形线圈) with many turns of conducting wire

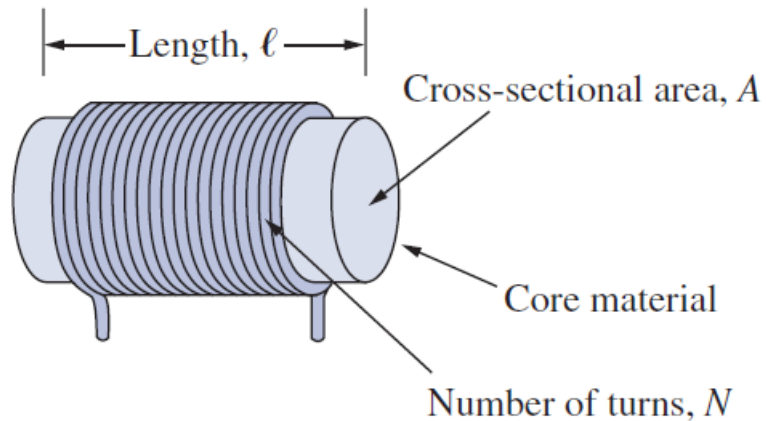
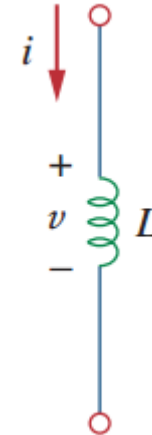


Figure 6.21

Typical form of an inductor. Permeability of the core, μ

$$v = L \frac{di}{dt}$$

$$L = \frac{N^2 \mu A}{\ell}$$



Joseph Henry (1797-1878)
American Physicist

- Inductance is the property that an inductor exhibits opposition to the change of current flowing through it, measured in **henrys** (H)

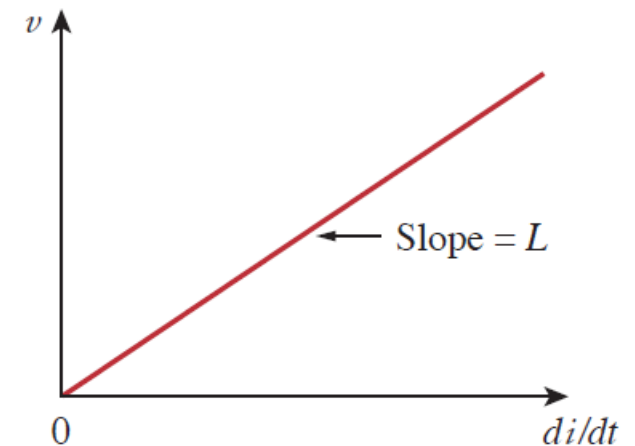
V-A Property of Inductors

- If the inductance of an inductor is independent of its current or voltage, then it is called a linear inductor. For linear inductor, the v-di/dt curve is a straight line
- The current-voltage relationship can be obtained as

$$v = L \frac{di}{dt} \rightarrow di = \frac{1}{L} v dt$$

- Integrating gives

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau \quad \text{Or} \quad i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$



- The power delivered to the inductor and the energy stored in it by increasing its current from 0 at t_0 to i at t are

$$p = vi = Li \frac{di}{dt} \rightarrow w = \int_{t_0}^t \left(Li' \frac{di'}{d\tau} \right) d\tau = \int_0^i Li' di' = \frac{1}{2} Li^2$$

Characteristic of Inductors

- An inductor acts like a **short circuit to dc** (对直流电压短路)
- The current through an inductor cannot change instantaneously (电流连续性)
- An ideal inductor does not dissipate energy (current remains constant, superconductor)
- A real inductor would have a small series **winding resistance** (线圈电阻) R_w , leading to a slow loss of the stored energy, and also a small parallel **winding capacitance** (线圈电容) C_w . Both R_w and C_w are small in value and can be ignorable in most applications (unless working at high frequency and/or high current conditions)

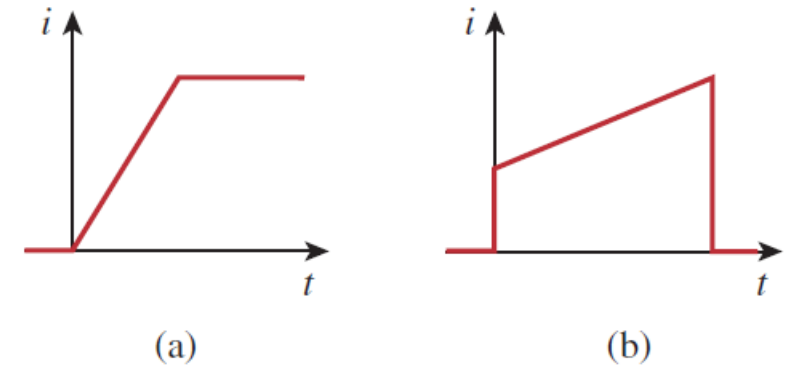


Figure 6.25

Current through an inductor: (a) allowed, (b) not allowable; an abrupt change is not possible.

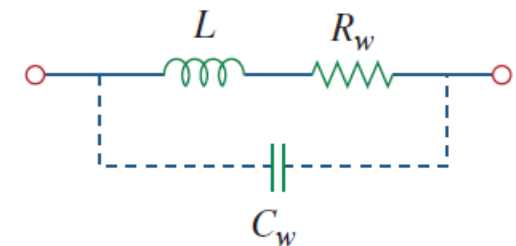


Figure 6.26

Circuit model for a practical inductor.

Example1: Inductors

The current through a 0.1-H inductor is $i(t) = 10te^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

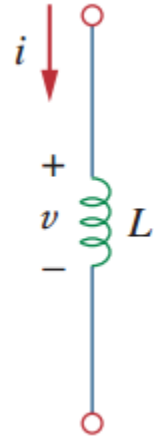
Solution:

Since $v = L di/dt$ and $L = 0.1$ H,

$$v = 0.1 \frac{d}{dt}(10te^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1 - 5t) \text{ V}$$

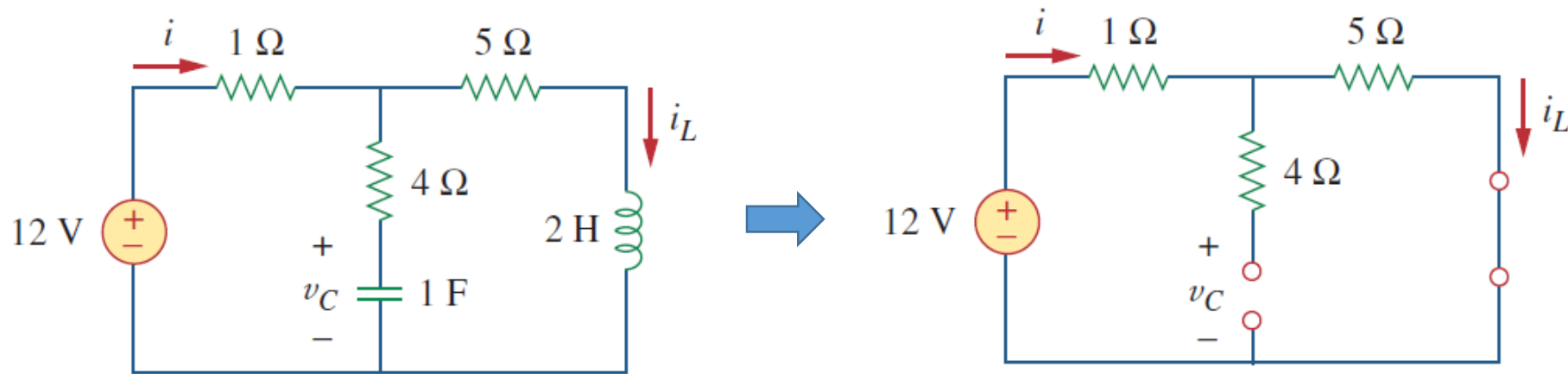
The energy stored is

$$w = \frac{1}{2}Li^2 = \frac{1}{2}(0.1)100t^2e^{-10t} = 5t^2e^{-10t} \text{ J}$$



Example2: Inductors

- Q: Consider the circuit in figure on the left. Under dc conditions, find: (a) i , v_C , and i_L , (b) the energy stored in the capacitor and inductor



Solution:

Capacitor is open-circuit and inductor is short-circuit for dc, so we can use the right figure to find the variables

(a) $v_C = 12 \cdot 5 / (1 + 5) = 10 \text{ V}$; $i_L = 12 / (1 + 5) = 2 \text{ A}$; $i = i_L = 2 \text{ A}$

(b) the energy stored in the capacitor is: $w = \frac{1}{2} \cdot C \cdot v_C^2 = \frac{1}{2} \cdot 1 \cdot 10^2 = 50 \text{ J}$

the energy stored in the inductor is: $w = \frac{1}{2} \cdot L \cdot i_L^2 = \frac{1}{2} \cdot 2 \cdot 2^2 = 4 \text{ J}$

Series Inductors

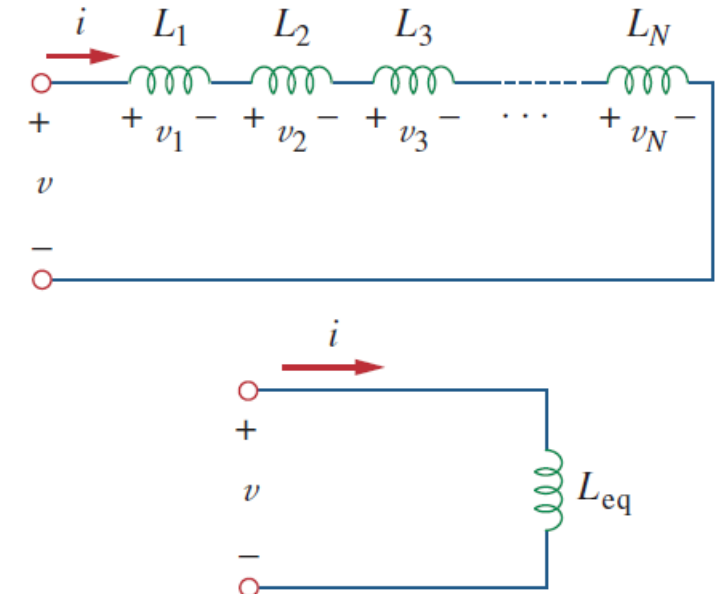
- For a series connection of N inductors, the same current flows through them. Applying KVL, we have

$$v = v_1 + v_2 + v_3 + \cdots + v_N$$

- Factoring in the voltage current relationship

$$\begin{aligned} v &= L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt} + \cdots + L_N \frac{di}{dt} \\ &= (L_1 + L_2 + L_3 + \cdots + L_N) \frac{di}{dt} = \left(\sum_{k=1}^N L_k \right) \frac{di}{dt} = L_{eq} \frac{di}{dt} \end{aligned}$$

$$\Rightarrow \boxed{L_{eq} = L_1 + L_2 + L_3 + \cdots + L_N}$$



- Thus, the equivalent inductance of series-connected inductors is the **sum of the individual inductances** → The same way as **resistors in series**

Parallel Inductors

- For a parallel connection of N inductors (并联电感), the voltage across them is the same. Applying KCL, we have

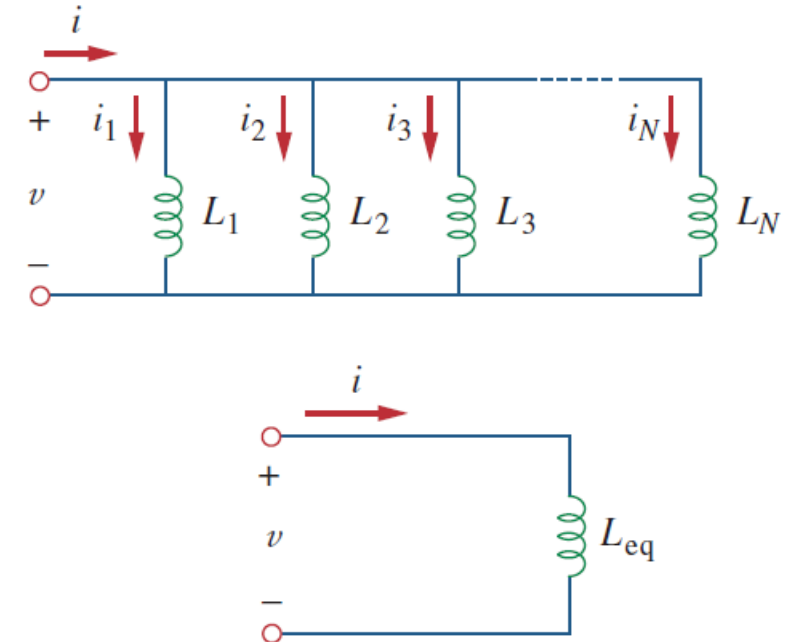
$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

- Factoring in the voltage current relationship

$$i = \left(\sum_{k=1}^N \frac{1}{L_k} \right) \int_{t_0}^t v \, dt + \sum_{k=1}^N i_k(t_0) = \frac{1}{L_{eq}} \int_{t_0}^t v \, dt + i(t_0)$$



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N}$$



- The equivalent inductance of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances → The parallel combination of inductors resembles that of resistors as well

Example1: Series and Parallel Inductors

- Q: Find the equivalent inductance of the circuit

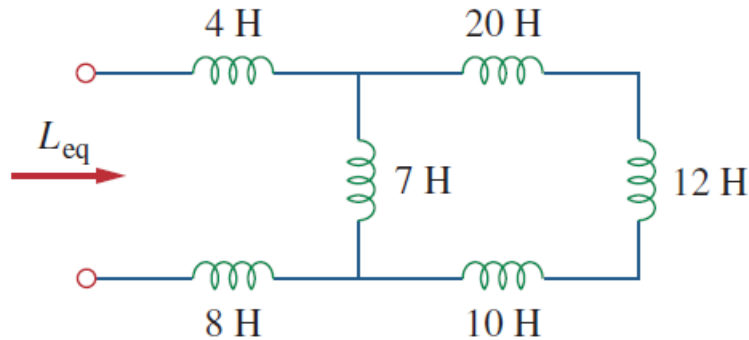


Figure 6.31

$$\begin{aligned}
 L_{eq} &= 4 + 8 + 7 \parallel (20 + 10) \\
 &= 12 + 7 \parallel 30 \\
 &= 12 + 6 \\
 &= 18 \text{ H}
 \end{aligned}$$

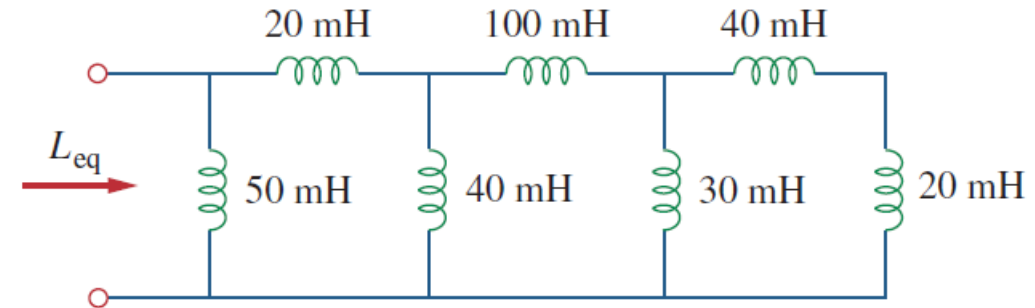


Figure 6.32

$$\begin{aligned}
 L_{eq} &= 50 \parallel (20 + 40 \parallel (100 + 30 \parallel (40 + 20))) \\
 &= 50 \parallel (20 + 40 \parallel (100 + 30 \parallel 60)) \\
 &= 50 \parallel (20 + 40 \parallel (100 + 20)) \\
 &= 50 \parallel (20 + 40 \parallel 120) \\
 &= 50 \parallel (20 + 30) \\
 &= 25 \text{ mH}
 \end{aligned}$$

Example2: Series and Parallel Inductors

Example 6.12 For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

Solution:

(a) Apply KCL at node n_1 :

$$i_1(0) = i(0) - i_2(0) = 4(2 - e^{-10 \cdot 0}) - (-1) = 5 \text{ mA}$$

(b) To find $v(t)$, we first calculate L_{eq}

$$L_{eq} = 2 + 4 \parallel 12 = 5 \text{ H}$$

$$v(t) = L_{eq} \frac{d}{dt} i(t) = 5 \frac{d}{dt} (4(2 - e^{-10t})) = 200e^{-10t} \text{ mV}$$

$$v_1(t) = 2 \frac{d}{dt} i(t) = 2 \frac{d}{dt} (4(2 - e^{-10t})) = 80e^{-10t} \text{ mV}$$

$$v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$$

(c) Currents are derived according to the v-i relation of inductor

$$i_1(t) = \frac{1}{4} \int_0^t v_2(t) dt + i_1(0) = 8 - 3e^{-10t} \text{ mA} \quad i_2(t) = \frac{1}{12} \int_0^t v_2(t) dt + i_2(0) = -e^{-10t} \text{ mA}$$

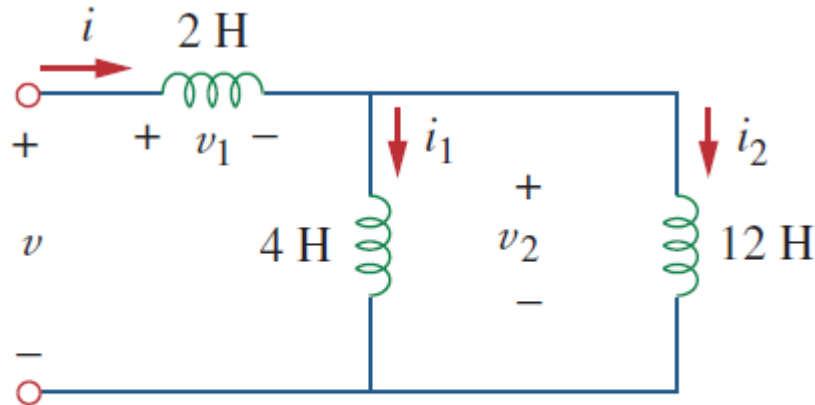


Figure 6.33

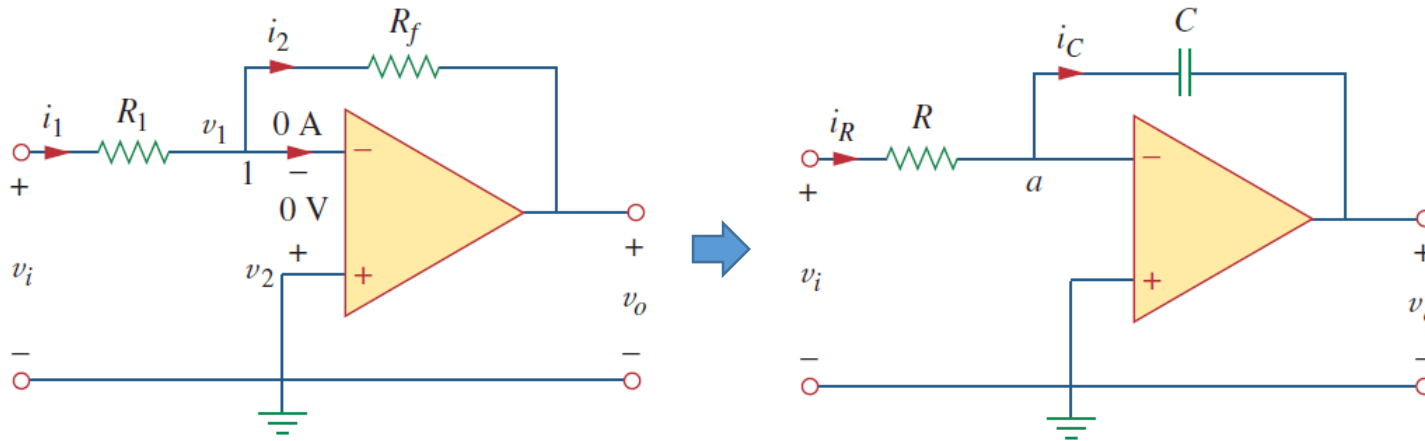
For Example 6.12.

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Application of Capacitors: Integrator

- Capacitors, in combination with op amps can be made to perform advanced mathematical functions such as the **integrator** (积分器)
- By replacing the feedback resistor with a capacitor, the output voltage of the inverting amplifier is a function of input's integration
 - In practice, the op amp integrator requires a feedback resistor to reduce dc gain and prevent saturation. Care must be taken that the op amp operates within the linear range



$$\frac{v_i}{R} = -C \frac{dv_o}{dt} \Rightarrow dv_o = -\frac{1}{RC} v_i dt$$

$$\Rightarrow v_o(t) - v_o(0) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$

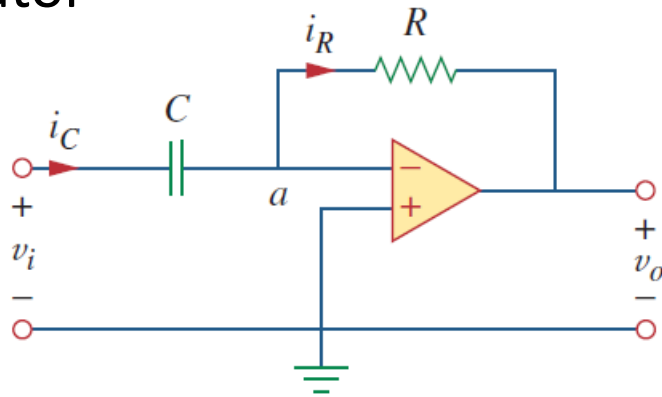
if $v_o(0) = 0$



$$v_o = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$

Application of Capacitors: Differentiator

- Another function is the **differentiator (微分器)**, whose output is proportional to the rate of change of the input signal
- By replacing the input resistor with a capacitor, the inverting amplifier is now a differentiator



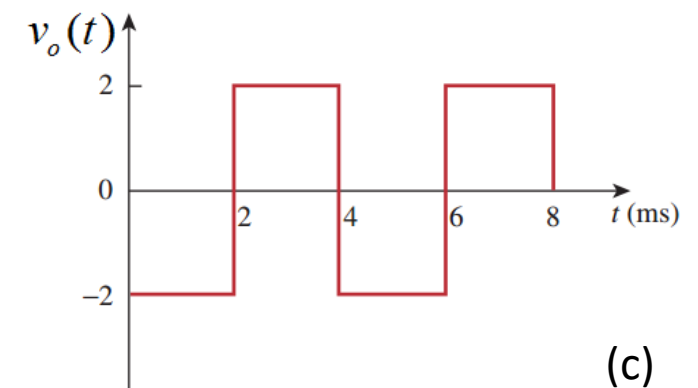
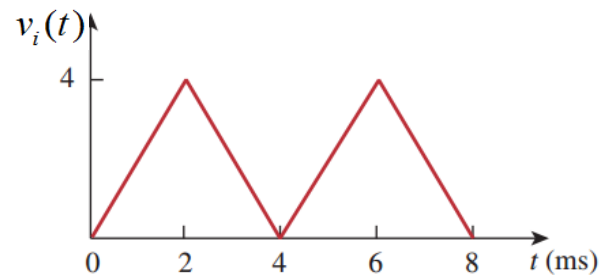
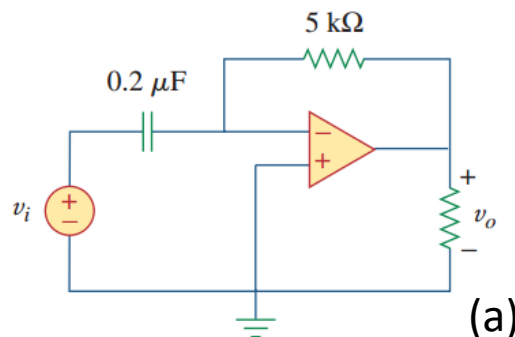
$$i_C = C \frac{dv_i}{dt}$$

$$i_R = -\frac{v_o}{R}$$



$$v_o = -RC \frac{dv_i}{dt}$$

- For example, in the following differentiator circuit of (a), if the input voltage has a waveform of (b), then its output will be as shown in (c)



Summary

- The current through a capacitor is directly proportional to the time rate of change of the voltage across it. A capacitor acts like an open circuit to a dc source, since no current will flow if the voltage is not changing
- The voltage across a capacitor is directly proportional to the time integral of the current through it. The voltage across a capacitor cannot change instantly
- Capacitors in series and in parallel are combined in the same way as conductance
- The voltage across an inductor is directly proportional to the time rate of change of the current through it. An inductor acts like a short circuit to a dc source, since the voltage will be zero if the current is not changing
- The current through an inductor is directly proportional to the time integral of the voltage across it. The current through an inductor cannot change instantly
- Inductors in series and in parallel are combined in the same way resistors in series and in parallel are combined
- At any given time, the energy stored in a capacitor is $\frac{1}{2}Cv^2$ while the energy stored in an inductor is $\frac{1}{2}Li^2$

Summary

TABLE 6.1

Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

[†] Passive sign convention is assumed.

Note: the [Delta-Wye](#) transformation can also be applied to inductors and capacitors in a similar manner, as long as all elements are the same type

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